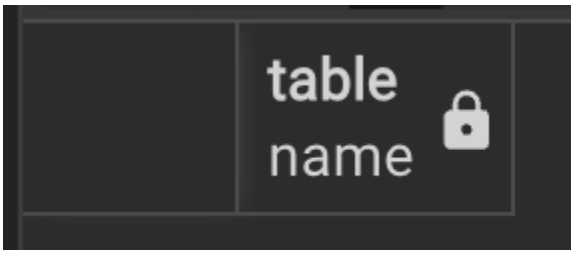
=== 1. Technical floor scorecard ===

	item text	need text	found text	status text	how_measured text
1	Tables	>= 6	13	PASS	base tables in the schema
2	Tables without a primary key	0	0	PASS	list in section 3a
3	Junction table for an M:N (composite key)	>= 1	3	PASS	key of 2+ columns, all of them foreign-key columns (heuristic)
4	Weak entity, ISA or recursive relationship	>= 1	weak 2, ISA 0, recursiv…	PASS	weak: key mixes FK and own columns; ISA: key is also an FK; recursive: FK to own table (heuri…
5	Foreign keys	>= 1 per related p…	17	PASS	ON DELETE policy for each is listed in section 4b
6	Foreign keys left at NO ACTION	each one delibera…	0	CHECK	NO ACTION is also what you get when you write nothing. Confirm you chose it (4b)
7	CHECK + UNIQUE (constraints and unique index…	>= 3	23	PASS	details in section 4a. Primary keys do not count
8	Views	>= 1	1	PASS	views and materialized views; details in section 5a
9	Triggers	>= 1	9	PASS	user triggers only; details in section 5b

-- === 2. Table roles read from the keys (heuristic) ===

	table name	primary_key text	fks_out bigint	fks_in bigint	looks_like text
1	ap_credit	transfer_credit_id	2	0	entity with FK(s)
2	course	course_id	0	6	strong entity (no F…
3	course_prerequisite	course_id, prereq_course…	2	0	junction (M:N)
4	enrollment	enrollment_id	2	0	entity with FK(s)
5	instructor	instructor_id	0	1	strong entity (no F…
6	major	major_id	0	2	strong entity (no F…
7	major_course_offeri…	major_id, course_id	2	0	junction (M:N)
8	section	course_id, term, section_…	2	2	weak entity?
9	student	student_id	0	6	strong entity (no F…
10	student_major	student_id, major_id	2	0	junction (M:N)
11	transfer_course	transfer_credit_id	2	0	entity with FK(s)
12	unit_override	student_id, term	1	0	weak entity?
13	waitlist	waitlist_id	2	0	entity with FK(s)

-- === 3a. Tables without a primary key (want none) ===



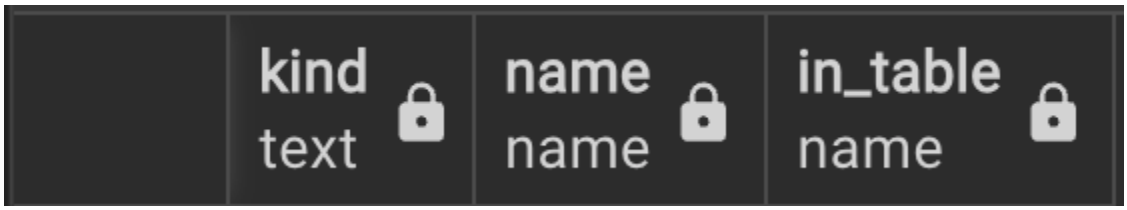
=== 3b. Foreign keys with no index starting on their columns ===

	table regclass	foreign_key name	columns text
1	section	section_instructor_id_fkey	instructor_id
2	enrollment	enrollment_course_id_term_section_no_fkey	course_id, term, section_...
3	transfer_course	transfer_course_course_id_fkey	course_id
4	transfer_course	transfer_course_student_id_fkey	student_id
5	ap_credit	ap_credit_course_id_fkey	course_id
6	ap_credit	ap_credit_student_id_fkey	student_id
7	major_course_offeri...	major_course_offering_course_id_fkey	course_id
8	student_major	student_major_major_id_fkey	major_id
9	course_prerequisite	course_prerequisite_prereq_course_id_fkey	prereq_course_id

=== 3c. Nullable foreign-key columns (compare with participation in the ER) ===

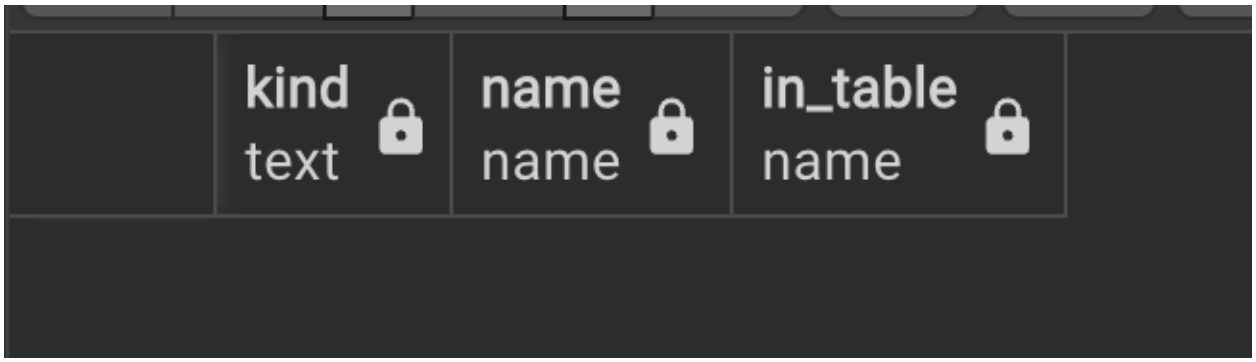
	table regclass	column name	references_table regclass
1	section	instructor_...	instructor
2	transfer_cour...	course_id	course
3	ap_credit	course_id	course

=== 3d. Table or column names that are reserved words (want none) ===



	kind text	name name	in_table name

-- === 3e. Names with capitals or spaces (pick one naming convention) ===



	kind text	name name	in_table name

-- === 3f. Columns that look like stored counts, totals or balances ===

	table name	column name	type text
1	ap_credit	transfer_credit…	integer
2	enrollment	time_enrolled	timestamp without time zo…
3	transfer_cour…	transfer_credit…	integer

-- === 4a. CHECK and UNIQUE rules ===

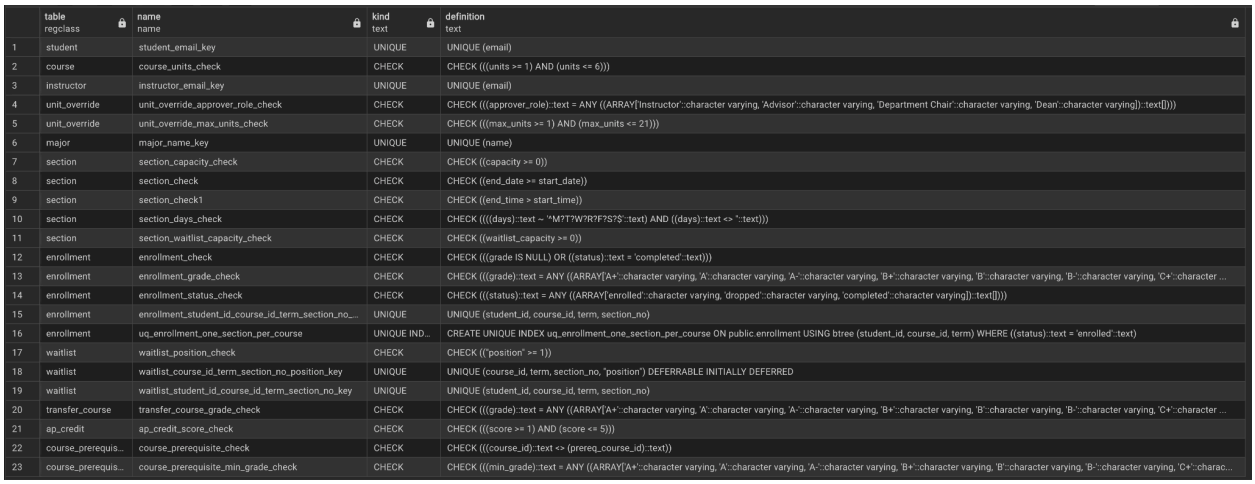


	table regclass	name name	kind text	definition text
1	student	student_email_key	UNIQUE	UNIQUE (email)
2	course	course_units_check	CHECK	CHECK (((units >= 1) AND (units <= 6)))
3	instructor	instructor_email_key	UNIQUE	UNIQUE (email)
4	unit_override	unit_override_approver_role_check	CHECK	CHECK (((approver_role)::text = ANY ((ARRAY['Instructor'::character varying, 'Advisor'::character varying, 'Department Chair'::character varying, 'Dean'::character varying])::text[])))
5	unit_override	unit_override_max_units_check	CHECK	CHECK (((max_units >= 1) AND (max_units <= 21)))
6	major	major_name_key	UNIQUE	UNIQUE (name)
7	section	section_capacity_check	CHECK	CHECK ((capacity >= 0))
8	section	section_check	CHECK	CHECK ((end_date >= start_date))
9	section	section_check1	CHECK	CHECK ((end_time > start_time))
10	section	section_days_check	CHECK	CHECK ((((days)::text ~ '^M?T?W?R?F?S?$'::text) AND ((days)::text <> ''::text)))
11	section	section_waitlist_capacity_check	CHECK	CHECK ((waitlist_capacity >= 0))
12	enrollment	enrollment_check	CHECK	CHECK (((grade IS NULL) OR ((status)::text = 'completed'::text)))
13	enrollment	enrollment_grade_check	CHECK	CHECK (((grade)::text = ANY ((ARRAY['A+'::character varying, 'A'::character varying, 'A-'::character varying, 'B+'::character varying, 'B'::character varying, 'B-'::character varying, 'C+'::character ...
14	enrollment	enrollment_status_check	CHECK	CHECK (((status)::text = ANY ((ARRAY['enrolled'::character varying, 'dropped'::character varying, 'completed'::character varying])::text[])))
15	enrollment	enrollment_student_id_course_id_term_section_no_...	UNIQUE	UNIQUE (student_id, course_id, term, section_no)
16	enrollment	uq_enrollment_one_section_per_course	UNIQUE IND...	CREATE UNIQUE INDEX uq_enrollment_one_section_per_course ON public.enrollment USING btree (student_id, course_id, term) WHERE ((status)::text = 'enrolled'::text)
17	waitlist	waitlist_position_check	CHECK	CHECK (("position" >= 1))
18	waitlist	waitlist_course_id_term_section_no_position_key	UNIQUE	UNIQUE (course_id, term, section_no, "position") DEFERRABLE INITIALLY DEFERRED
19	waitlist	waitlist_student_id_course_id_term_section_no_key	UNIQUE	UNIQUE (student_id, course_id, term, section_no)
20	transfer_course	transfer_course_grade_check	CHECK	CHECK (((grade)::text = ANY ((ARRAY['A+'::character varying, 'A'::character varying, 'A-'::character varying, 'B+'::character varying, 'B'::character varying, 'B-'::character varying, 'C+'::character ...
21	ap_credit	ap_credit_score_check	CHECK	CHECK (((score >= 1) AND (score <= 5)))
22	course_prerequis...	course_prerequisite_check	CHECK	CHECK (((course_id)::text <> (prereq_course_id)::text))
23	course_prerequis...	course_prerequisite_min_grade_check	CHECK	CHECK ((((min_grade)::text = ANY ((ARRAY['A+'::character varying, 'A'::character varying, 'A-'::character varying, 'B+'::character varying, 'B'::character varying, 'B-'::character varying, 'C+'::charac...

-- === 4b. Foreign keys and their ON DELETE / ON UPDATE policy ===

	table regclass	foreign_key name	references_table regclass	on_delete text	on_update text	deferrable boolean
1	unit_override	unit_override_student_id_fkey	student	CASCADE	NO ACTION	false
2	section	section_course_id_fkey	course	CASCADE	NO ACTION	false
3	section	section_instructor_id_fkey	instructor	SET NULL	NO ACTION	false
4	enrollment	enrollment_course_id_term_section_no_fkey	section	CASCADE	NO ACTION	false
5	enrollment	enrollment_student_id_fkey	student	CASCADE	NO ACTION	false
6	waitlist	waitlist_course_id_term_section_no_fkey	section	CASCADE	NO ACTION	false
7	waitlist	waitlist_student_id_fkey	student	CASCADE	NO ACTION	false
8	transfer_course	transfer_course_course_id_fkey	course	SET NULL	NO ACTION	false
9	transfer_course	transfer_course_student_id_fkey	student	CASCADE	NO ACTION	false
10	ap_credit	ap_credit_course_id_fkey	course	SET NULL	NO ACTION	false
11	ap_credit	ap_credit_student_id_fkey	student	CASCADE	NO ACTION	false
12	major_course_offeri...	major_course_offering_course_id_fkey	course	CASCADE	NO ACTION	false
13	major_course_offeri...	major_course_offering_major_id_fkey	major	CASCADE	NO ACTION	false
14	student_major	student_major_major_id_fkey	major	RESTRICT	NO ACTION	false
15	student_major	student_major_student_id_fkey	student	CASCADE	NO ACTION	false
16	course_prerequisite	course_prerequisite_course_id_fkey	course	CASCADE	NO ACTION	false
17	course_prerequisite	course_prerequisite_prereq_course_id_fkey	course	RESTRICT	NO ACTION	false

=== 4c. Columns ===

	table name	# smallint	column name	type text	nullability text	default text
1	ap_credit	1	transfer_credit…	integer	NOT NULL	[null]
2	ap_credit	2	score	integer	NOT NULL	[null]
3	ap_credit	3	student_id	integer	NOT NULL	[null]
4	ap_credit	4	course_id	character varying(10)		[null]
5	course	1	course_id	character varying(10)	NOT NULL	[null]
6	course	2	name	character varying(100)	NOT NULL	[null]
7	course	3	units	integer	NOT NULL	[null]
8	course	4	department	character varying(50)	NOT NULL	[null]
9	course_prerequisite	1	course_id	character varying(10)	NOT NULL	[null]
10	course_prerequisite	2	prereq_course_…	character varying(10)	NOT NULL	[null]
11	course_prerequisite	3	min_grade	character varying(2)	NOT NULL	'C-'::character varying
12	course_prerequisite	4	override	boolean	NOT NULL	false
13	enrollment	1	enrollment_id	integer	NOT NULL	[null]
14	enrollment	2	student_id	integer	NOT NULL	[null]
15	enrollment	3	course_id	character varying(10)	NOT NULL	[null]
16	enrollment	4	term	character varying(100)	NOT NULL	[null]
17	enrollment	5	section_no	integer	NOT NULL	[null]
18	enrollment	6	status	character varying(10)	NOT NULL	'enrolled'::character varying
19	enrollment	7	grade	character varying(2)		[null]
20	enrollment	8	time_enrolled	timestamp without time zo…	NOT NULL	CURRENT_TIMESTAMP
21	instructor	1	instructor_id	integer	NOT NULL	[null]
22	instructor	2	first_name	character varying(50)	NOT NULL	[null]
23	instructor	3	last_name	character varying(50)	NOT NULL	[null]
24	instructor	4	email	character varying(100)	NOT NULL	[null]
25	major	1	major_id	integer	NOT NULL	[null]
26	major	2	name	character varying(100)	NOT NULL	[null]
27	major_course_offeri…	1	major_id	integer	NOT NULL	[null]
28	major_course_offeri…	2	course_id	character varying(10)	NOT NULL	[null]
29	section	1	course_id	character varying(10)	NOT NULL	[null]
30	section	2	term	character varying(100)	NOT NULL	[null]
31	section	3	section_no	integer	NOT NULL	[null]
32	section	4	start_date	date	NOT NULL	[null]
33	section	5	end_date	date	NOT NULL	[null]
34	section	6	capacity	integer	NOT NULL	[null]
35	section	7	waitlist_capacity	integer	NOT NULL	0

36	section	8	days	character varying(7)	NOT NULL	[null]
37	section	9	start_time	time without time zone	NOT NULL	[null]
38	section	10	end_time	time without time zone	NOT NULL	[null]
39	section	11	location	character varying(50)		[null]
40	section	12	instructor_id	integer		[null]
41	student	1	student_id	integer	NOT NULL	[null]
42	student	2	first_name	character varying(50)	NOT NULL	[null]
43	student	3	last_name	character varying(50)	NOT NULL	[null]
44	student	4	address	character varying(200)		[null]
45	student	5	phone_number	character varying(20)		[null]
46	student	6	email	character varying(100)	NOT NULL	[null]
47	student_major	1	student_id	integer	NOT NULL	[null]
48	student_major	2	major_id	integer	NOT NULL	[null]
49	transfer_course	1	transfer_credit…	integer	NOT NULL	[null]
50	transfer_course	2	grade	character varying(2)	NOT NULL	[null]
51	transfer_course	3	student_id	integer	NOT NULL	[null]
52	transfer_course	4	course_id	character varying(10)		[null]
53	unit_override	1	student_id	integer	NOT NULL	[null]
54	unit_override	2	term	character varying(100)	NOT NULL	[null]
55	unit_override	3	max_units	integer	NOT NULL	[null]
56	unit_override	4	approver_name	character varying(100)	NOT NULL	[null]
57	unit_override	5	approver_role	character varying(20)	NOT NULL	[null]
58	unit_override	6	approved_date	date	NOT NULL	CURRENT_DATE
59	waitlist	1	waitlist_id	integer	NOT NULL	[null]
60	waitlist	2	student_id	integer	NOT NULL	[null]
61	waitlist	3	course_id	character varying(10)	NOT NULL	[null]
62	waitlist	4	term	character varying(100)	NOT NULL	[null]
63	waitlist	5	section_no	integer	NOT NULL	[null]
64	waitlist	6	position	integer	NOT NULL	[null]
65	waitlist	7	join_date	timestamp without time zo…	NOT NULL	CURRENT_TIMESTAMP

-- === 5a. Views ===

	view name	updatable character varying (3)	check_option character varying	definition_start text
1	section_seat_stat…	NO	NONE	SELECT s.course_id, s.term, s.section_no, c.name AS course_name, COALESCE((((i.first_name)::text || ' '::text) || (i.la

-- === 5b. Triggers ===

	table regclass	trigger name	timing text	events text	level text	function regproc
1	section	trg_section_instructor	BEFORE	INSERT UPDATE	ROW	check_instructor_schedule
2	enrollment	trg_enrollment_capacity	BEFORE	INSERT UPDATE	ROW	check_section_capacity
3	enrollment	trg_enrollment_leave_waitl…	AFTER	INSERT UPDATE	ROW	leave_waitlist_on_enroll
4	enrollment	trg_enrollment_prereqs	BEFORE	INSERT UPDATE	ROW	check_prereqs
5	enrollment	trg_enrollment_schedule	BEFORE	INSERT UPDATE	ROW	check_student_schedule
6	enrollment	trg_enrollment_units	BEFORE	INSERT UPDATE	ROW	check_unit_limit
7	enrollment	trg_waitlist_promote	AFTER	UPDATE	ROW	promote_from_waitlist
8	waitlist	trg_waitlist_capacity	BEFORE	INSERT UPDATE	ROW	check_waitlist_capacity
9	waitlist	trg_waitlist_not_enrolled	BEFORE	INSERT UPDATE	ROW	check_waitlist_not_enroll…

-- === 5c. Trigger functions: do they mention FOR UPDATE / LOCK? ===

	function regproc	has_for_update boolean	has_lock_table boolean	counts_rows boolean
1	check_section_capacity	true	false	true
2	check_waitlist_capacity	true	false	true
3	check_unit_limit	true	false	false
4	check_prereqs	false	false	false
5	check_student_schedule	false	false	false
6	check_instructor_schedule	false	false	false
7	check_waitlist_not_enrolled	false	false	false
8	leave_waitlist_on_enroll	false	false	false
9	promote_from_waitlist	false	false	false

-- === 6a. Table size and row counts ===

	table name	exact_rows bigint	est_rows bigint	dead_rows bigint	table_size text	index_size text	total_size text	last_analyze timestamp with time zone	last_autoanalyze timestamp with time zone
1	enrollment	7	7	0	8192 bytes	48 kB	56 kB	[null]	[null]
2	waitlist	3	3	0	8192 bytes	48 kB	56 kB	[null]	[null]
3	major	3	3	0	8192 bytes	32 kB	40 kB	[null]	[null]
4	instructor	3	3	0	8192 bytes	32 kB	40 kB	[null]	[null]
5	student	5	5	0	8192 bytes	32 kB	40 kB	[null]	[null]
6	student_major	6	6	0	8192 bytes	16 kB	24 kB	[null]	[null]
7	course	5	5	0	8192 bytes	16 kB	24 kB	[null]	[null]
8	transfer_course	2	2	0	8192 bytes	16 kB	24 kB	[null]	[null]
9	section	6	6	0	8192 bytes	16 kB	24 kB	[null]	[null]
10	major_course_offeri…	8	8	0	8192 bytes	16 kB	24 kB	[null]	[null]
11	ap_credit	2	2	0	8192 bytes	16 kB	24 kB	[null]	[null]
12	unit_override	1	1	0	8192 bytes	16 kB	24 kB	[null]	[null]
13	course_prerequisite	4	4	0	8192 bytes	16 kB	24 kB	[null]	[null]

-- === 6b. Reads and writes per table ===

	table name	seq_scan bigint	idx_scan bigint	inserts bigint	updates bigint	deletes bigint	pct_index_scans numeric
1	course	2	30	5	0	0	93.8
2	student	3	25	5	0	0	89.3
3	section	2	22	6	0	0	91.7
4	enrollment	4	15	7	0	0	78.9
5	major	3	14	3	0	0	82.4
6	waitlist	4	7	3	0	0	63.6
7	instructor	3	5	3	0	0	62.5
8	unit_override	2	4	1	0	0	66.7
9	course_prerequisite	2	4	4	0	0	66.7
10	student_major	2	0	6	0	0	0.0
11	transfer_course	2	0	2	0	0	0.0
12	ap_credit	2	0	2	0	0	0.0
13	major_course_offeri…	2	0	8	0	0	0.0

-- === 6c. Indexes ===

	table name	index name	times_used bigint	size text	kind text
1	ap_credit	ap_credit_pkey	0	16 kB	primary key
2	course	course_pkey	30	16 kB	primary key
3	course_prerequisite	course_prerequisite_pkey	4	16 kB	primary key
4	enrollment	enrollment_pkey	0	16 kB	primary key
5	enrollment	enrollment_student_id_course_id_term_section_no_...	0	16 kB	unique
6	enrollment	uq_enrollment_one_section_per_course	15	16 kB	unique
7	instructor	instructor_email_key	0	16 kB	unique
8	instructor	instructor_pkey	5	16 kB	primary key
9	major	major_name_key	0	16 kB	unique
10	major	major_pkey	14	16 kB	primary key
11	major_course_offeri...	major_course_offering_pkey	0	16 kB	primary key
12	section	section_pkey	22	16 kB	primary key
13	student	student_email_key	0	16 kB	unique
14	student	student_pkey	25	16 kB	primary key
15	student_major	student_major_pkey	0	16 kB	primary key
16	transfer_course	transfer_course_pkey	0	16 kB	primary key
17	unit_override	unit_override_pkey	4	16 kB	primary key
18	waitlist	waitlist_course_id_term_section_no_position_key	7	16 kB	unique
19	waitlist	waitlist_pkey	0	16 kB	primary key
20	waitlist	waitlist_student_id_course_id_term_section_no_key	0	16 kB	unique

-- === 6d. Buffer cache hits ===

	table name	disk_reads bigint	cache_hits bigint	pct_hit numeric
1	course	0	61	100.0
2	student	0	55	100.0
3	section	0	55	100.0
4	major	0	31	100.0
5	enrollment	0	29	100.0
6	major_course_offeri…	0	24	100.0
7	student_major	0	18	100.0
8	instructor	0	13	100.0
9	course_prerequisite	0	12	100.0
10	waitlist	0	11	100.0
11	ap_credit	0	6	100.0
12	transfer_course	0	6	100.0
13	unit_override	0	2	100.0

-- === 6e. Column statistics (n_distinct < 0 means a fraction of the row count

	table name	column name	null_frac real	n_distinct real	avg_width integer
1	ap_credit	course_id	0	-1	6
2	ap_credit	score	0	-1	4
3	ap_credit	student_id	0	-1	4
4	ap_credit	transfer_credit…	0	-1	4
5	course	course_id	0	-1	6
6	course	department	0	-0.4	16
7	course	name	0	-1	31
8	course	units	0	-0.4	4
9	course_prerequisite	course_id	0	-0.75	6
10	course_prerequisite	min_grade	0	-0.25	3
11	course_prerequisite	override	0	-0.5	1
12	course_prerequisite	prereq_course_…	0	-1	6
13	enrollment	course_id	0	-0.5714286	6
14	enrollment	enrollment_id	0	-1	4
15	enrollment	grade	0.71428573	-0.28571427	2
16	enrollment	section_no	0	-0.42857143	4
17	enrollment	status	0	-0.42857143	9
18	enrollment	student_id	0	-0.71428573	4
19	enrollment	term	0	-0.2857143	10
20	enrollment	time_enrolled	0	-0.14285715	8
21	instructor	email	0	-1	20
22	instructor	first_name	0	-1	5
23	instructor	instructor_id	0	-1	4
24	instructor	last_name	0	-1	5
25	major	major_id	0	-1	4
26	major	name	0	-1	16
27	major_course_offeri…	course_id	0	-0.625	6
28	major_course_offeri…	major_id	0	-0.375	4
29	section	capacity	0	-0.6666667	4
30	section	course_id	0	-0.6666667	6
31	section	days	0	-0.5	2
32	section	end_date	0	-0.33333334	4
33	section	end_time	0	-0.6666667	8
34	section	instructor_id	0.16666667	-0.5	4
35	section	location	0	-0.8333333	7

36	section	section_no	0	-0.5	4
37	section	start_date	0	-0.33333334	4
38	section	start_time	0	-0.6666667	8
39	section	term	0	-0.33333334	10
40	section	waitlist_capacity	0	-0.5	4
41	student	address	0	-1	33
42	student	email	0	-1	19
43	student	first_name	0	-1	5
44	student	last_name	0	-1	5
45	student	phone_number	0	-1	13
46	student	student_id	0	-1	4
47	student_major	major_id	0	-0.5	4
48	student_major	student_id	0	-0.8333333	4
49	transfer_course	course_id	0	-1	6
50	transfer_course	grade	0	-1	2
51	transfer_course	student_id	0	-1	4
52	transfer_course	transfer_credit…	0	-1	4
53	unit_override	approved_date	0	-1	4
54	unit_override	approver_name	0	-1	14
55	unit_override	approver_role	0	-1	5
56	unit_override	max_units	0	-1	4
57	unit_override	student_id	0	-1	4
58	unit_override	term	0	-1	10
59	waitlist	course_id	0	-0.33333334	7
60	waitlist	join_date	0	-0.33333334	8
61	waitlist	position	0	-1	4
62	waitlist	section_no	0	-0.33333334	4
63	waitlist	student_id	0	-1	4
64	waitlist	term	0	-0.33333334	10
65	waitlist	waitlist_id	0	-1	4

=== 7. FD check: does term determine start_date in Section? (0 rows = no counterexample)===

